

BMS_SLAVE

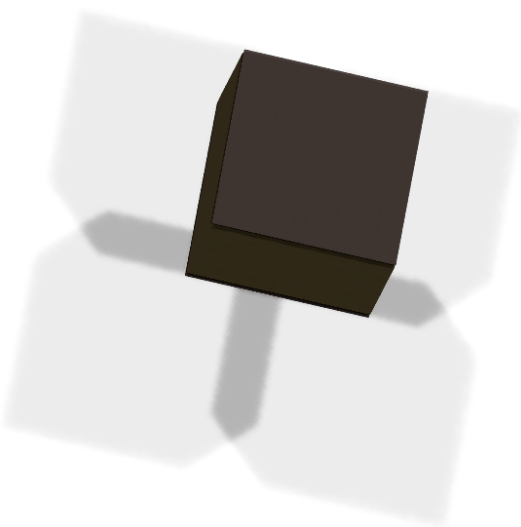
Variant: DRAFT

2026-09-24

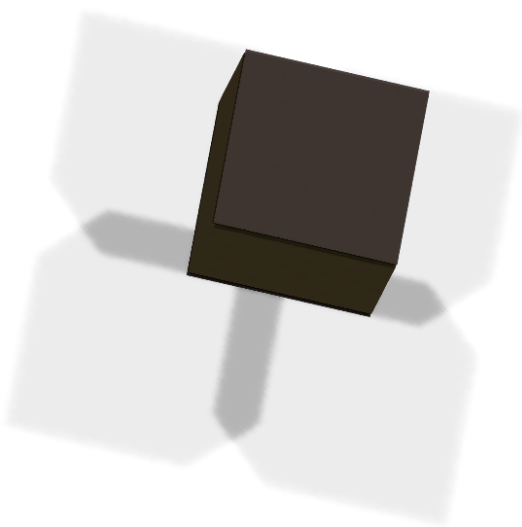
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 24-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

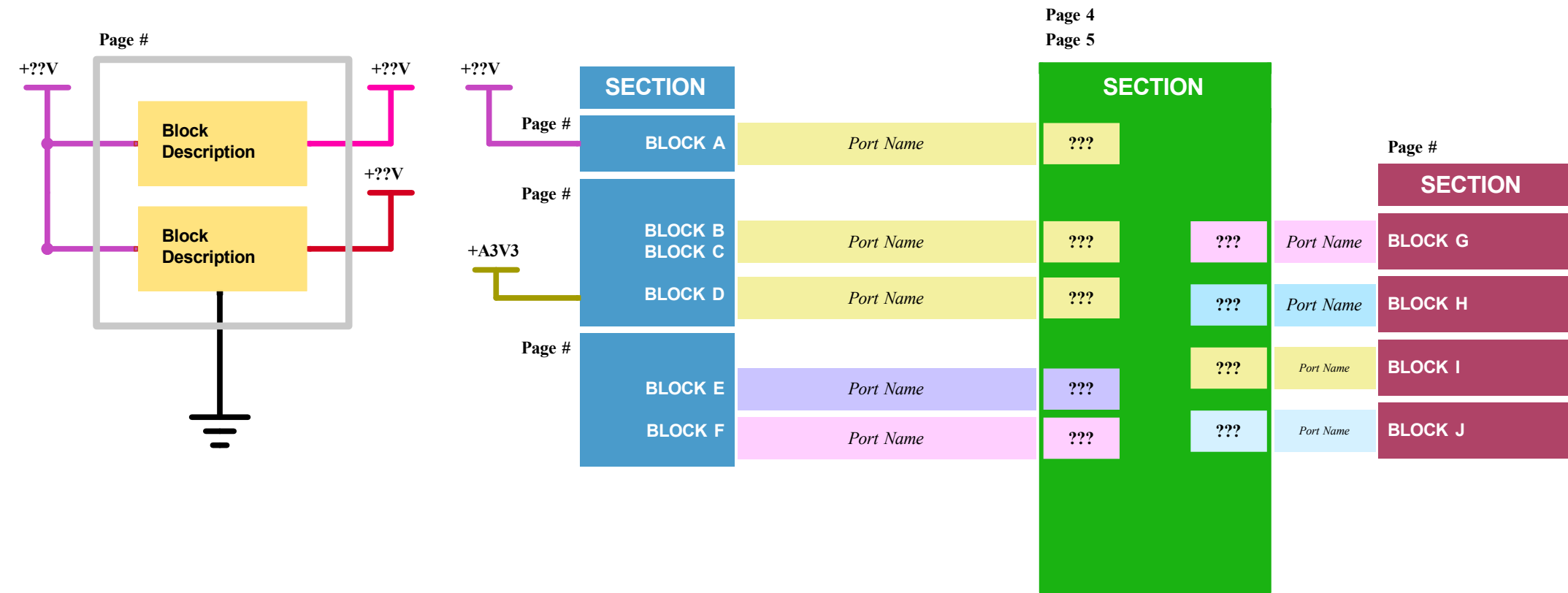
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:		Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
	Sheet Title:		Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Path: /		File Name: bms_slave.kicad_sch		Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
		Reviewer:		Size: A3	Sheet: 1 of 10	

[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??



Comments:	Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 10

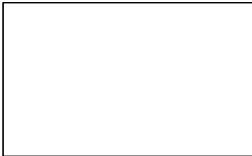
[3] Project Architecture

Page 4

Connectors



Cell Sense & Balance



Power & Reference



Temperature Sense



Daisy Chain



Mechanical



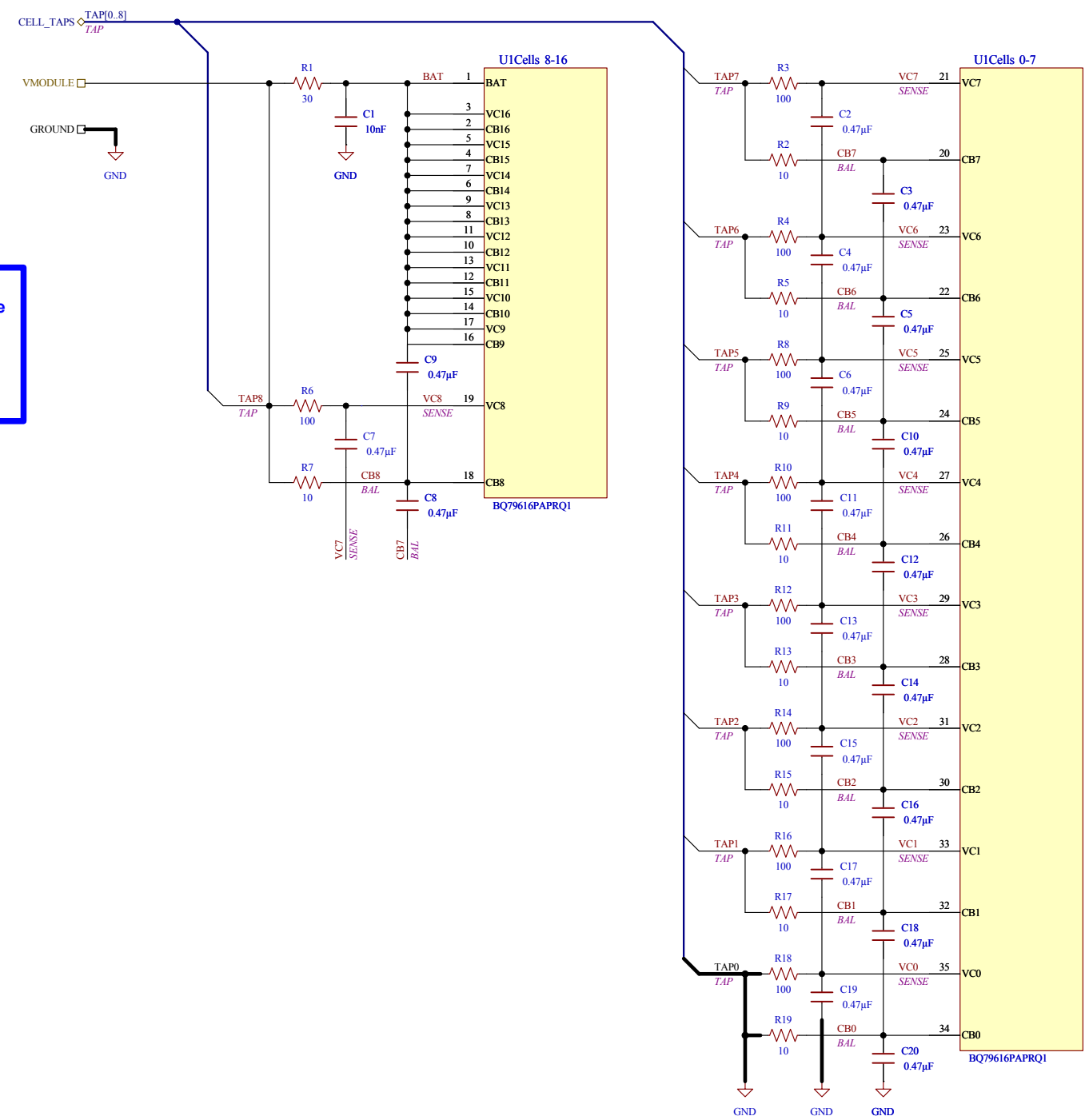
	Comments:		Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
			Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/			Reviewer:	Size: A3	Sheet: 3 of 10	

[4] Connectors

	Comments:		Company:		Variant: DRAFT	Git Hash: 87d29ad
			Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Title: Connectors	File Name: connectors.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:	
	Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A4	Sheet: 4 of 10	

[5] Cell Sense & Balance

LAYOUT NOTE:
Open/NC pins should have
minimum trace lengths
and must not be
connected to a wire or
cable



Comments:	Company:		Variant:	Git Hash:
			DRAFT	87d29ad
Sheet Title:	Board Name:		Project Name:	
	Cell Sense & Balance		BMS Slave	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/Cell Sense & Balance/	Aidan Brzezinski		
			Size:	Sheet:
			A3	5 of 10

[6] Power & Reference

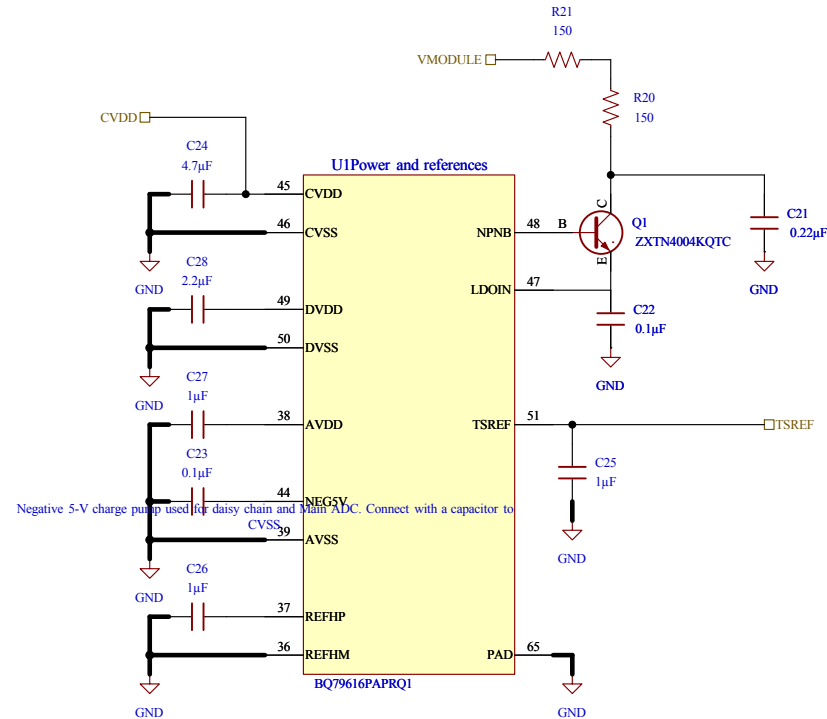
LAYOUT NOTE: U1 GROUNDING (TI SLUSE81F §10.1.1)

All grounds are one net (GND) in the schematic. In layout, each bypass cap returns to its own ground pin through a small L1 copper island. Islands join the L2 GND plane through vias only. Do NOT connect islands to each other on L1.

GROUND	PIN	CAPS RETURNING TO IT
AVSS	39	AVDD (38), BAT filter, VCO filter, CB0 filter, thermal pad (65)
DVSS	50	DVDD (49)
CVSS	46	LDOIN (47), NEG5V (44), CVDD (45), TSREF (51), GPIO filter caps, daisy-chain bypass caps
REFHM	36	REFHP (37) ONLY - keep separate from AVSS on L1, join at L2 plane

RULES

- Cap within 2 mm of its pin, same layer, no via between cap and pin.
- Cap ground pad -> its island -> via to L2. Not a shared trace.
- Keep pins 40-43 (COMH/COML) clear, place caps above and below the daisy-chain exit, not in it.
- Thermal pad (65): 36 vias to L2 GND plane, tented/plugged.



Comments:

Company:

Variant:

Git Hash:

DRAFT

87d29ad

Board Name:

BMS_SLAVE

Project Name:

BMS Slave

Sheet Title:

Power & Reference

File Name:

power_and_reference.kicad_sch

Designer:

Aidan Brzezinski

Date:

Revision:

Sheet Path:

/Project Architecture/Power & Reference/

Reviewer:

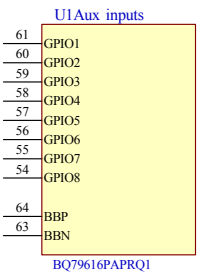
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A4

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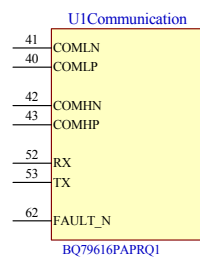
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[7] Temperature Sense



Comments:	Company:		Variant:	Git Hash:
	Board Name: BMS_SLAVE		DRAFT	87d29ad
Sheet Title: Temperature Sense	File Name: temperature_sense.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Temperature Sense/		Reviewer:	Size: A4	Sheet: 7 of 10

[9] Daisy Chain



Comments:	Company:		Variant:	Git Hash:
	Board Name: BMS_SLAVE		DRAFT	87d29ad
Sheet Title: Daisy Chain	File Name: daisy_chain.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Daisy Chain/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Mechanical

	Comments:		Company:		Variant: DRAFT	Git Hash: 87d29ad
			Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Title: Mechanical	File Name: mechanical.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:	
	Sheet Path: /Project Architecture/Mechanical/		Reviewer:	Size: A4	Sheet: 10 of 10	

[11] Revision History



Comments:	Company: Blue Devil Motor Sports		Variant: DRAFT	Git Hash: 87d29ad
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 11 of 10